Xingyue Liu

aroma@mail.dlut.edu.cn | 15842668147 | recynie.github.io
Github | Google Scholar

Profile

Dalian University of Technology Dalian, China

Artificial Intelligence, School of Future Technology

2023-2027 Undergraduate

Academic Score: 87.0/100
Language Test: CET-4 618, CET-6 591

Publications

1. **Xingyue Liu**†, Zijie. Xing†, Runze. Wang, Luoming. Hu, and Yanming. Shen*, "Language-guided expertise evolution for protein optimization," *ICLR 2026 Workshop RSI. (Accepted)* [Link]
2. R. Wang, Y. Ma, **Xingyue Liu**, Z. Xing, and Y. Shen*, "Motif-driven molecular graph representation learning," *Expert Systems with Applications*, vol. 269, p. 126484, 2025. *(Published)* [Link]
3. Runze Wang, Zijie Xing, **Xingyue Liu**, Mingqi Yang, Che He, Yanming Shen*, "From Graphs to Tokens: Substructure-Aware Molecular Representation for Large Language Models," *Information Processing and Management. (Published)* [Link]
4. Zijie Xing†, **Xingyue Liu**†, Runze Wang, Luoming Hu, Yanming Shen*, "KnowEvo: Knowledge Evolution for Protein Optimization," *NeurIPS 2026. (Accepted)* [PDF]

Research Experience

04/2025 | 05/2026

Language-Guided Expertise Evolution for Protein Optimization

Advisor: Prof. Yanming Shen

Overview: Developed a self-evolving agent system for protein optimization through explore–verify–refine cycles, accumulating executable prior knowledge in a code-based knowledge base.

Core Contributions:

- Code as Knowledge: Represented knowledge as a code population, with gating mechanisms to match and optimize target protein sub-distributions.
- Search Algorithm: Used Thompson sampling for training and LCB scoring for final knowledge selection, balancing exploration and exploitation.
- Knowledge Tree Updates: Built an agent workspace with parent-node strategies, validation results, and metrics to support feedback-driven exploration and analysis.
- Self-Evolving System: Built a high-concurrency agent system from scratch, decoupling exploration from knowledge updates and supporting multiple servers for efficient resource use.

Outcome: An early version won the Best Innovation Award at the Bohr+SciMaster AI4S Agent Development Competition (only 2 teams awarded) and was shortlisted for the Qizhi Startup Camp interview (top 10%). Published as co-first author at ICLR 2026 RSI Workshop [1] and NeurIPS 2026 [4].

10/2024 – 12/2024 | **Molecular Representation Learning**

Advisor: Prof. Yanming Shen

Overview: Integrated molecular motifs into GNNs and LLMs to improve representation learning beyond the limits of 1-WL testing and 1D SMILES.

Core Contributions:

- Uni-Motif Framework (GNNs): Designed and evaluated three substructure feature fusion mechanisms and compared motif decomposition strategies.
- S^2 Token Strategy (LLMs): Proposed and evaluated substructure-aware tokenization; reproduced leading baselines to validate motif-augmented performance gains.

Outcome: Published as co-author in two peer-reviewed journals: *Expert Systems with Applications* [2] and *Information Processing and Management* [3].

12/2023 – Present | **Prediction of Protein Crystallization Conditions**

Advisor: Prof. Song Xue

Overview: Predicted protein crystallization conditions by modeling the relationships among protein sequences, structures, and chemical environments.

Core Contributions:

- Dataset Construction: Built a crystallization condition database (~7k proteins), extracting, filtering, and integrating crystallization records from the PDB.
- Crystallization Condition Similarity Analysis: Used Foldseek to cluster structurally similar proteins and analyze solution-component combinations and crystallization condition distributions.
- pLM Fine-tuning: Fine-tuned SaProt (a pretrained structure-aware protein language model) on the crystallization condition task.

Outcome: Found that pLMs struggle to represent crystallization conditions. Current work adds hydration as a downstream modality to improve performance and develops hydration-aware protein representations.

Awards & Honors

1. AY 2023–2024 **Academic Excellence Scholarship**
2. AY 2024–2025 **Academic Excellence Scholarship**
3. AY 2024–2025 **National Scholarship**

Research Interests

- **Auto Scientific Research**: Improve the long-horizon scientific research ability of AI, focusing on iterating ideas with experimental feedback to propose valuable, feasible ideas.
- **Protein Representation**: Develop structure-aware and hydration-aware protein representations to improve their performance on downstream tasks.

Skills

- **AI agent**: Self-evolving agents, agent harness design
- **AI for Science**: Protein representation learning, molecular representation learning, protein optimization